Lightweight U-Net Variants for Medical Image Segmentation

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Abstract

Medical image segmentation needs accurate models that run on small clinical hardware. We design lightweight U-Net variants and evaluate them on a public cardiac MRI dataset.

Keywords: medical image segmentation, U-Net, cardiac MRI, lightweight models

1. Introduction

Hospitals often lack powerful GPUs. We ask how much a U-Net can be shrunk before segmentation quality drops.

2. Related Work

U-Net and its variants are standard for biomedical segmentation. Compression of such networks is less explored.

3. Methodology

We reduce channel widths and replace standard convolutions with depthwise separable ones. Models are trained on cardiac MRI slices with a Dice loss.

4. Results

The smallest variant kept a Dice score of 0.89 with 8 times fewer parameters than the full U-Net.

5. Limitations

Only one cardiac dataset was used. Results may not generalise to other scanners.

6. Conclusion

Compact U-Net variants can approach full-size accuracy.

7. Future Work

Future work should test other organs and scanner types.

References

[1] O. Ronneberger et al. style U-Net references are discussed generally in the text.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.